

Solution Requirements

Date	01 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The application allows users to access the OptiCrop website and use the crop recommendation system without complex authentication.	Medium
2	Authorization levels	All users are permitted to enter soil and environmental parameters and receive crop recommendations..	Low
3	External interfaces	The system interacts with the Flask web application, HTML/CSS frontend, and the trained machine learning model (.pkl).	High

4	Transactions processing	The system accepts soil and weather inputs, processes them using the trained ML model, and predicts the most suitable crop.	High
5	Reporting	The application displays the recommended crop based on the prediction result.	High
6	Business rules, etc.	Crop prediction is based on Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall. Invalid inputs are rejected.	High
7	Compliance to laws or regulations	The application follows standard software development practices and securely processes user inputs without permanently storing personal information.	Medium
8	<i>Other</i>	The system provides a simple, responsive, and user-friendly interface for quick crop prediction.	Medium

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	The system should generate crop recommendations quickly after receiving valid user inputs.	<i>Prediction time less than 2 seconds</i>
2	Scalability	The application should support multiple users with minimal performance degradation.	<i>Supports multiple concurrent users</i>
3	Security & Data Privacy	User inputs should be processed securely without permanently storing personal information.	<i>No sensitive data stored permanently</i>
4	Reliability & Availability	The application should provide consistent and accurate crop recommendations and remain available during normal operation.	<i>99% application availability</i>
5	Usability & Accessibility	The interface should be simple, responsive, and easy for farmers and other users to understand.	<i>User-friendly interface with clear navigation</i>
6	<i>Other</i>	The application should have modular, reusable, and well-documented code for easy updates and future improvements.	Modular code structure and documentation

